

Machine learning approach for detection of flooding DoS attacks in 802.11 networks and attacker localization

Mayank Agarwal · Dileep Pasumarthi ·
Santosh Biswas · Sukumar Nandi

Received: 17 May 2014 / Accepted: 16 October 2014 / Published online: 2 November 2014
© Springer-Verlag Berlin Heidelberg 2014

Abstract IEEE 802.11 Wi-Fi networks are prone to a large number of Denial of Service (DoS) attacks due to vulnerabilities at the media access control (MAC) layer of 802.11 protocol. In this work, we focus on the flooding DoS attacks in Wi-Fi networks. In flooding DoS attacks, a large number of legitimate looking spoofed requests are transmitted to a victim access point (AP). The processing of large number of spoofed frames results in a huge load at the AP, resulting in a flooding DoS attack. Current methods to detect the flooding DoS use encryption, signal characteristics, protocol modification, upgradation to newer standards etc. which are often expensive to operate and maintain. In this paper, we propose a novel Machine Learning (ML) based intrusion detection system along with intrusion prevention system (IPS) that not only detects the flooding DoS attacks in Wi-Fi networks, but also helps the victim station (STA) in recovering swiftly from the attack. To the best of our knowledge, the usage of ML based techniques for detection of flooding DoS attacks in 802.11 networks has largely been unexplored. The ML based IDS detects the flooding DoS attacks with a high accuracy (precision) and detection rate (recall). After the attack is detected, the location of the attacker is ascertained using

Angle of Arrival based localization algorithm and traffic coming from the attacker region is blocked which helps in mitigating the effect of flooding DoS attack.

Keywords 802.11 · Flooding DoS attacks · Wi-Fi networks · Intrusion detection system · Machine learning · Sniffer · Localization

1 Introduction

IEEE 802.11 Wireless Local Area Networks (WLANs) [9] have become increasingly popular for the past few years. Millions of wireless Access Points (APs) have been deployed across the world enabling the users to stay connected while on the move. Wi-Fi offers significant advantages in terms of deployment costs, ease of installation, expansion and roaming benefits. In the areas where wired networks cannot be deployed due to difficult terrain, Wi-Fi networks prove to be an effective solution.

In pursuit of offering increasing user benefits, security features in Wi-Fi networks are often overlooked. Since wireless transmissions travel over the air, an eavesdropper just needs to be within the wireless range of the target AP in order to sniff the frames traveling in the network. An attacker possessing a Wi-Fi capable device along with penetrating operating system like BackTrack [3] can easily launch a myriad of attacks on a Wi-Fi network.

IEEE initially proposed Wired Equivalent Privacy (WEP) as primary encryption technique for secure communication between wireless hosts. However many loop holes were discovered in WEP implementation. The works in [14, 36–38] and tools [1, 7] have shown that WEP could be easily broken. The inadequacy of WEP to provide a robust encryption scheme led to the development of the

M. Agarwal · D. Pasumarthi · S. Biswas (✉) · S. Nandi
Department of Computer Science and Engineering, Indian
Institute of Technology, Guwahati 781039, India
e-mail: santosh_biswas@iitg.ernet.in;
santoshbiswas402@yahoo.com

M. Agarwal
e-mail: mayank.agl@iitg.ernet.in

D. Pasumarthi
e-mail: p.dileep@iitg.ernet.in

S. Nandi
e-mail: sukumar@iitg.ernet.in

802.11i standard. The 802.11i standard proposed Wi-Fi Protected Access (WPA) and WPA2 as standard encryption techniques which provided significant advantages over WEP in terms of encryption. User authentication which is absent in WEP, is made possible in WPA and WPA2 through the Extensible Authentication Protocol (EAP). However, it may be noted that the different encryption techniques like WEP, WPA, WPA2 etc. only encrypt the data frames. The management and control frames are transmitted in clear-text. 802.11 management frames are responsible for establishment and maintenance of communications in Wi-Fi network(s). Management frames are used to support authentication, association and synchronization. 802.11 control frames aid in the delivery of data frames between STAs. A majority of the attacks that take place in 802.11 exploit the clear-text and un-authenticated nature of the management and control frames. 802.11w [10] is a recent standard proposed by IEEE which has provisions for encryption of management frames like de-authentication and dis-association frames. However they are not enough to prevent the flooding Denial of Service (DoS) attacks in 802.11 networks.

In this paper, we focus on the Authentication and Association flooding DoS attacks in 802.11 networks. An authentication flooding DoS attack occurs when an adversary inundates the victim AP with large number of spoofed authentication request frames. The AP cannot handle the barrage of spoofed authentication request frames, and eventually becomes un-responsive to handle the legitimate requests. On similar lines, an adversary can flood the AP with large number of spoofed Association Request frames. Both these attacks result in DoS on the AP, thereby forbidding the genuine users from accessing the services offered by the AP.

Current methods to deal with flooding DoS attacks are based on:

- Encryption.
- Protocol modification.
- Up-gradation to newer standards.
- Use of Received signal strength indicator (RSSI) characteristics.

These approaches have associated drawbacks with them. Encryption requires key management, key distribution and certificate management which involve heavy computation and adds to administrative overhead. Upgradation to newer standard is a costly process and is not always possible due to the presence of legacy networks and the associated cost. RSSI requires use of specialized hardware equipment in order to work correctly. Thus, we see that adoption of the existing schemes leads to increased running as well as maintenance costs.

Machine learning and soft computing techniques like fuzzy logic, artificial neural networks are being widely

used for various practical applications in different domains like hydro projects, particle swarm optimization, model parameter optimization etc. [16, 33, 34, 39, 40, 42]. In our work, we propose a Machine Learning based intrusion detection system (IDS) for the detection of flooding DoS attacks in 802.11 networks. The proposed scheme does not suffer from the drawbacks listed above. The results obtained for detection of flooding DoS attacks using machine learning based IDS are quite promising with both the accuracy and detection rate crossing 95 % mark. Our proposed IDS also incorporates an additional module that locates the attacker and blocks traffic coming from the attacker region in order to minimize the impact of flooding DoS attacks. Angle of Arrival (AoA) and RSSI based approaches can be used for locating the attacker. In our localization module we have used AoA based approach since it performs better than RSSI based localization. To the best of our knowledge, none of the approaches in the literature use machine learning based methods to detect flooding DoS attacks in 802.11 Wi-Fi networks.

The summary of the contributions are:

1. We propose a Machine Learning based IDS that not only helps in detecting the existence of flooding based DoS attacks in a Wi-Fi network but also helps the AP to recover from the attack quickly. The proposed IDS strictly adheres to the 802.11 standard and does not require any protocol modification. The only hardware requirement is a sensor capable of sniffing the wireless data. This ensures that the technique is cost-effective.
2. Our approach can be applied to both open as well as encrypted networks. Since the proposed technique does not alter any protocol, it can be deployed to new as well as legacy Wi-Fi networks. It does not require patching of underlying operating system, application and is independent of the client software.
3. We also propose an additional localization algorithm which estimates location of the adversary. After locating the attacker, frames from the attacker region are discarded. We have utilized RSSI and AoA based localization approaches for locating the attacker (not machine learning). Finally, we adopt the AoA based approach since it performs better than RSSI based approach.

Our paper is organized as follows. In Sect. 2 we discuss the basics of 802.11 communication along with authentication and association flooding DoS attacks. A detailed study of existing approaches to tackle the authentication and association flooding DoS attacks are also presented in Sect. 2. We describe our proposed Machine Learning based IDS along with the Localization algorithm in Sect. 3. The design of the ML based IDS is elaborated in Sect. 4. The results for accuracy, detection rate and localization for the

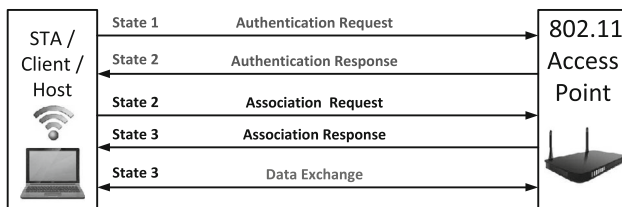


Fig. 1 4-way handshake between STA and access point

proposed Machine Learning based Intrusion Detection are elaborated in Sect. 5. Finally we conclude our paper in Sect. 6.

2 Background and motivation

In this section, we look at the basics of authentication and association in 802.11 networks and the vulnerabilities associated with them. The authentication and association flooding DoS attacks are also elaborated. We also discuss the existing approaches to tackle the authentication and association flooding DoS attacks and their drawbacks. Finally we describe the motivation behind our work.

Authentication and Association Flooding DoS attacks are severe, since they overload the AP with spoofed authentication and association frames resulting in the degradation or complete stalling of the services offered by the AP. An attacker need not be authenticated nor associated with an AP to launch these attacks. The attacker can launch these attacks simultaneously on multiple STAs with minimal resources.

Figure 1 shows the 4-way handshake that takes place between a station (STA)¹ and an access point (AP) before a STA can access the services offered by the AP. It also depicts the various states a STA traverses before it can start exchanging data with the AP. A Wi-Fi compliant STA can be in one of the three states (State 1, State 2, State 3) at any given time.

A STA is in State 1 at the beginning. A STA is neither authenticated nor associated in State 1. In State 1, the STA sends Authentication Request frame. In an open (un-encrypted) Wi-Fi network, this frame just consists of the MAC address of the STA. In encrypted Wi-Fi networks the STA sends the encryption key along with its MAC address. Upon receiving the Authentication Request frame the AP sends an Authentication Response frame indicating a successful or unsuccessful authentication. If the Authentication Response is a failure the connection is dropped. If the Authentication Response is successful the STA moves to State 2. A STA is authenticated but not associated in State 2. To

access the services offered by the AP the STA needs to associate with the AP. The STA sends an Association Request frame to the AP. The AP replies with an Association Response frame and the STA associates with the AP. The data exchange between STA and AP takes place after successful association. The STA enters in State 3, which indicates that the STA is successfully authenticated as well as successfully associated.

2.1 Vulnerabilities of management and control frames in 802.11 networks

After the introduction of 802.11i standard in 2004, Wireless Local Area Network (WLAN) devices provide robust encryption and authentication features. The anomalies of WEP were eliminated with the advent of WPA and WPA2. The 802.11i also uses the Advanced Encryption Standard (AES) to guarantee confidentiality and integrity of the data transferred between the communicating hosts. However the major drawback in any encryption scheme of 802.11 is that they only encrypt the data frames. The management and control frames still travel un-encrypted. The management and control frames are important frames which facilitate connection establishment, maintenance and termination of STA with AP. The authentication and association frames are management frames and are sent in plain-text. The main motivation behind sending of these frames in plain-text are speedy processing and low computation for the AP. Spoofing plain-text frames is a trivial job for an attacker. Since these frames are sent in plain-text there is no way of authenticating them and the AP is left with no choice but to process the spoofed frames sent by the malicious users. We now describe the Authentication and Association Flooding DoS attacks in detail.

2.2 Authentication and association flooding DoS attacks

1. *Authentication request flooding DoS attack* Authentication enables an AP to ascertain the identity of a STA before the STA is allowed to associate. In authentication flooding DoS attack the attacker sends a large number authentication request frames with falsified MAC addresses. APs can handle limited number of authentication request(s) at a given time. If a large number of authentication frames are directed towards an AP, the AP gets stuck in processing the enormous amount of authentication frames. This results in wastage of AP's resources and eventually the AP is unable to provide services to genuine STAs. If the AP uses some encryption mechanisms like WEP, WPA or WPA2, the number of authentication frames required to stall the AP are even less [13], since encryption

¹ In this paper the terms STA, Client, Host have been used interchangeably.

techniques involve heavy computation and a small number of forged authentication frames can induce a heavy workload on the AP, resulting in DoS attack. Figure 2 shows the Authentication Flooding DoS attack. In Step 1 of Fig. 2, which is a pre-attack scenario, the clients (Users 1 and 2) normally authenticate to AP and continue their data exchange with the AP. In Step 2, the attacker inundates the AP with large number of authentication frames with falsified MAC addresses. Since authentication frames are un-authenticated, AP has no mechanism to verify them. An AP can handle only a limited number of connections at any given time. The performance of AP degrades drastically due to handling of the large number of requests and subsequently the AP is not in a position to provide services to any client due to overloading of spoofed authentication request frames. In Step 3, when a legitimate user wishes to authenticate to AP the request is rejected since the AP needs to first recover from the Authentication Flooding DoS attack in order to serve the client. It is equally likely that the attacker may be present in the network much before the network becomes functional. In such a scenario the attacker launches the flooding DoS attacks as soon as the network becomes operative. In this case, the Step 1 shown in Fig. 2 is skipped and only Step 2 and Step 3 suffice to launch the flooding DoS attacks. To further elaborate the authentication flood attack we refer to authentication frame format shown in Fig. 3. The entries shown in Bold in the authentication frame format represent the possible fields that can be manipulated by the attacker for launching authentication flooding attack which are explained below.

- *Address 2 field (STA MAC)* This is the MAC address of the STA that is communicating with the AP. Attacker can manipulate the address so that it appears to come from a legitimate station.
- *Sequence control field* This is a 12 bit field that increments by 1 for every frame that is transmitted. Since it's a 12 bit field it can take any of the possible $2^{12} = 4,096$ values. When a STA authenticates it can choose any random value as the starting sequence number. The start value varies with the implementation. The sequence number can take any arbitrary value under normal and attack scenarios.
- *FCS—Frame check sequence* This field computes the checksum of the frame which is useful for the checking its integrity. If the re-computed checksum at the destination does not match with the source frame checksum the frame is dropped. So the

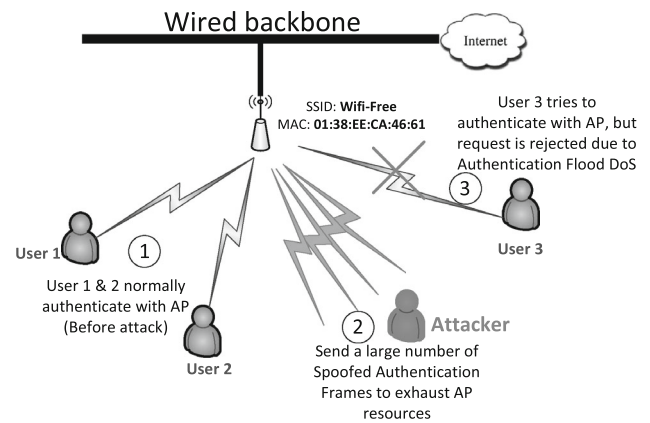


Fig. 2 Authentication flooding DoS attack

checksum value varies depending on the values of various fields of the frame.

Thus, we see that all the three visible differentiating parameters in the authentication frame format are inadequate for discriminating between normal and attack, whether used individually or in combination. Hence for detection of flooding DoS attack a machine learning based IDS is required which can combine various frame statistics in order to differentiate between normal and attack scenarios.

2. *Association flooding DoS attack* In association flooding DoS attack the attacker overflows the AP with large number of association request frames having fraudulent MAC addresses. The AP maintains an association table which consists of the MAC addresses of the users and their Association IDs (AIDs). However, the size varies depending upon the capabilities of the AP. Sending of large number of spoofed association frames results in the association table of the AP getting full, and the AP can no longer provide service to new client(s). The structure for association request frame is similar to authentication frame shown in Fig. 3 except for the SubType field which is set to 01 and ToDS bit which is set to 1.

The attacker can use a number of freely available tools like *aircrack-ng* suite [2], *file2air* [5], *scapy* [6] to launch flooding DoS attacks. An attacker requires to have a list of spoofed MAC addresses, Basic Service Set Identifier (i.e., MAC address of AP), Service Set Identifier (SSID) of the network and the channel number on which the AP is running in order to launch the flooding DoS attacks. These information are sent in clear-text and can be easily extracted using tools like Wireshark [8], *airodump-ng*, *tcpdump*, *kismet* etc. To make the flooding DoS attacks

Fig. 3 Authentication frame format for a legitimate STA**802.11 MAC Header**

Frame Control 0xB000	Duration ID 258 μsec	Address 1 AP MAC	Address 2 STA MAC	Address 3 BSSID (AP) MAC	Sequence Control	Address 4 Unused	Network Data	FCS Checksum		
Protocol Version 0	Type 0	Sub-Type 11	ToDS 0	FromDS 0	More Frag 0	Retry 0	Power Mgmt 0	More Data 0	WEP 0	Order 0

stealthier and evade detection the attacker can insert random silence periods between sending of attack frame(s).

2.3 Existing approaches to deal with authentication and association flooding DoS attacks

In this sub-section, we look at the approaches proposed in the literature to deal with Authentication and Association DoS attacks.

1. *Encryption based methods* Bellardo et al. in [12] proposed modification of the 802.11 authentication framework. Bellardo suggested that if all the frames are authenticated an attacker would not be able to launch the flooding DoS attacks. However, authenticating all 802.11 frames would require upgradation of both the client as well as the AP firmware. The authentication of all 802.11 frames would generate an immense load on the AP, which might lead to DoS on AP. The 802.11w standard drafted in 2009 has provision for encrypting de-authentication and disassociation frames. However, it requires firmware upgrades on both AP as well as STA. The existence of large number of legacy Wi-Fi devices proves to be a major hindrance for the adoption of 802.11w standard. Though 802.11w proposes encryption for certain management frames, however, probe request and authentication frames are still sent in clear-text. Thus, in case of authentication and association flooding DoS attacks switching to 802.11w standard is futile.
2. *Received signal strength indicator (RSSI) based methods* Ivan et al. [32] proposed a methodology for detection of flooding DoS attacks wherein they partition the entire Wi-Fi area into regions. They make use of broadcast communication, signal propagation, and dense deployment of IEEE 802.11 technology in-order to obtain the regions. The motive behind this is, once the attacker region is identified all frames coming from that particular region are ignored. Though this method can help locate a stationary attacker it would fail if the attacker is mobile. Also the method requires the clients to invest time in determining its neighbors. The clients are also required to send their neighbor list in association request frame. This requires firmware upgradation and cannot be deployed to legacy networks. Faria and Cheriton [19] have made use of signal print techniques in order to overcome flooding DoS attacks. Their core idea is that signal print generated from a particular STA helps in identifying the devices more accurately as compared to other identifying parameters like MAC address. It is obvious that STAs located close to each other will produce similar signal prints which may obscure the identification of the malicious device.
3. *Sequence number based methods* Mao et al. [23], Mar et al. [31], Wright [4], Xia et al. [41] and Anjum et al. [11] have suggested different schemes for detection of spoofing attacks based on the sequence number analysis. Sequence number is incremented by 1 in each frame. If the STA sends a frame with sequence number 'x' the successive frame is sent with the sequence number 'x+1' and so on. If the next frame is sent with a sequence number greater than 'x+1' it is quite likely that the frame is spoofed because the actual sequence must have been 'x+1' under normal network circumstances. Though an intelligent attacker can predict the sequence number in advance to escape detection, sending a frame with sequence number 'x+1' at the precise timing is often difficult if the number of frames to be sent are high.
4. *Centralized methods* A centralized framework like 802.1X equipped with EAP can help effectively block the flooding DoS attacks. This is because in 802.1X systems all the communications are handled by the Authenticator. A client (known as Supplicant in the 802.1X terminology) cannot access the network services directly without successfully authenticating to the Authenticator. 802.1X provides various authenticating means like user-name/password, tunneling, certificates etc., which ensures that only authentic users connect to the network. However, such measures suffer from single

point of failure. The compromise of the main authentication server may lead to additional delays in offering services to users. If the attack on the main authentication server is severe the whole system would stop thereby affecting all the services offered by the AP.

5. *Neural networks and genetic programming based methods* Liu et al. [28, 29] have used neural network based IDS for detection of data link layer attack in Wi-Fi networks. However their proposed IDS suffer from high false alarm rate. LaRoche et al. [25, 26] propose use of genetic programming for detection of de-authentication attacks but does not have any mechanism for detection of flooding DoS attack.

To summarize, the drawbacks of the current approaches to detect and prevent the flooding DoS attacks are as follows:

1. Requires alteration in 802.11 protocol to support Authentication and Encryption of management frames which are currently non-authenticated.
2. Expensive setup.
3. Upgradation to newer standards.
4. Requires specialized hardware instruments.

From the above points we can conclude that an effective flooding based DoS detection technique is required to have the following features:

1. Should not require modification of 802.11 protocol stack.
2. Ensure ease of installation on legacy as well as new Wi-Fi networks.
3. Hardware costs should not be prohibitive.
4. Should not depend on the client's underlying operating system, application etc.
5. Non Cryptography approach to ensure light-weight scheme.
6. Locate the attacker after the flooding DoS attack is detected and discard the frames coming from the attacker.

We now discuss our proposed ML based IDS that embodies the features listed above and overcomes the drawbacks of the existing approaches.

3 Proposed machine learning based IDS

The proposed architecture of the ML based IDS is shown in Fig. 4. The IDS consists of the following main components.

3.1 Knowledge base

The dataset generated is stored in the *Knowledge Base*. The process of dataset generation is described in Sect. 4. The generated dataset may contain noisy data called outliers.

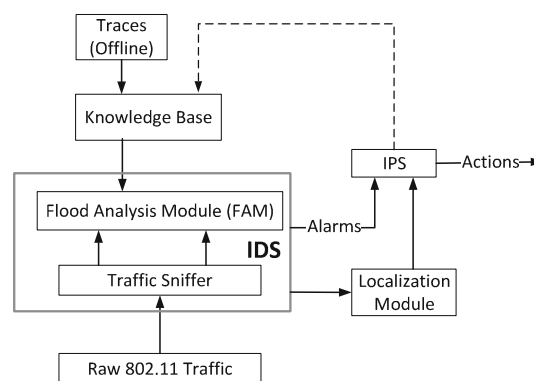


Fig. 4 IDS architecture

Outliers are those observations whose probability of occurrence is extremely small. Removal of such outliers is needed for robustness in data analysis and helps the IDS to achieve better accuracy. The following pre-processing is done on the dataset to exclude outliers:

- Since the sniffer works in promiscuous mode, it captures all the frames within its vicinity. This also includes frames directed to different APs other than those being monitored. Frames directed to different APs are dropped.
- Few STA have only sent Probe Request and Probe Response frames to the AP(s). These STA never authenticated with the AP. The Knowledge Base component leaves out processing of such STAs.

The *Knowledge Base* component also updates itself regularly by getting the feedback from the Intrusion Prevention System (IPS) module.

3.2 Intrusion detection system (IDS)

The *Intrusion detection system* is the main module that determines the occurrence of the flooding DoS attacks. This module consists of two sub-modules: Traffic Sniffer and the Flood Analysis modules, which are explained next.

- *Traffic sniffer* The *Traffic Sniffer* is a wireless sniffer running in promiscuous mode which captures the raw network traffic pertaining to the monitored AP. Frames to other APs are ignored. The Traffic Sniffer passes the frames collected to the Flood Analysis Module for further investigation.
- *Flood analysis module (FAM)* The classification of the network traffic takes place in *Flood Analysis Module*. The *Knowledge Base* component after pre-processing submits the traces to the *Flood Analysis Module*. The *Flood Analysis Module* is trained using these traces. The training process is performed off-line. After training the FAM the IDS is deployed on the live

network. The FAM analyzes the live network frames obtained from the Traffic Sniffer. The sniffer collects all the relevant frames for each STA that is associated with the AP being monitored. From the moment a STA sends an Authentication Request frame till the time it leaves the network the FAM module monitors the data exchange activities of each STA. While capturing the network statistics for various STAs the FAM determines whether flooding DoS attacks have occurred or not. If the flooding DoS attacks have indeed occurred, the FAM informs the Intrusion Prevention System (IPS) as well as the Localization module. The Localization module determines the location of the adversary node and informs the AP. The AP blocks the traffic coming from the attacker region only.

The details regarding test bed generation, feature selection, the various classifiers used for the ML based IDS are explained in the Sect. 4.

3.3 Localization module

The Localization module is an add-on that helps to locate the attacker in the network. After the attacker is located, the AP blocks the traffic coming from the attacker region till the flooding DoS attacks subsides. Localization algorithms generally make use of RSSI and AoA in-order to locate nodes (not machine learning). We first look at an RSSI based approach followed by AoA approach.

3.3.1 RSSI based localization

Liu et al. [27] use RSSI based approach along with a voting based scheme for localization. In this approach they divide the target field into grid cells as shown in Fig. 5 and locator nodes (L_i) determine how likely the node to be located (N_j , say) is in each grid cell. Using the RSSI of the received frames, a locator node L_i estimates the distance of node N_j as r_j . The approach then creates a concentric virtual ring using the distance $r_j + \delta$ and $r_j - \delta$ with L_i as center, where δ is the error in estimation. Voting is done in this concentric virtual ring and the section receiving the highest votes is declared to be the region of the node N_j . A region has a vote count of 1 by default. If two regions overlap the vote count increases to 2 and if three regions overlap the vote count increases to 3. For ‘ k ’ overlapping regions, the vote count becomes ‘ k ’. A cell(s) with the highest vote is chosen and the “centroid” of the cell(s) is the estimated location of the node N_j . In the example shown in Fig. 5, L_1 , L_2 and L_3 are used for locating the node N_1 . The probable location of N_1 is the centroid of the the grid cells having a vote count of 3.

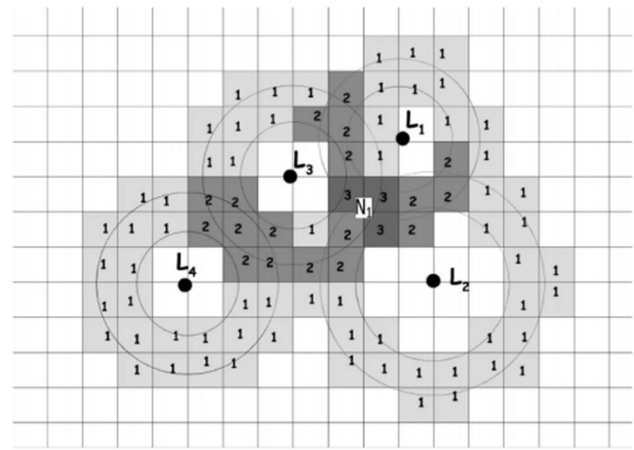


Fig. 5 RSSI based localization

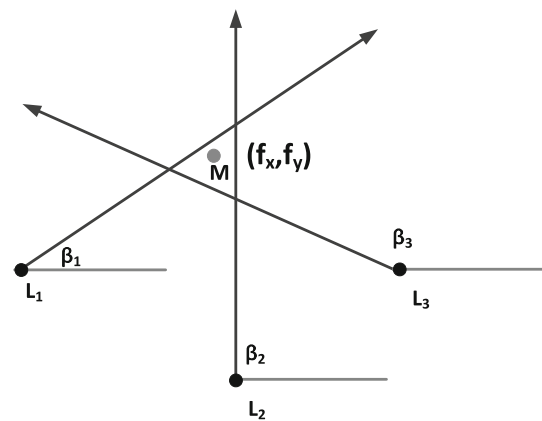


Fig. 6 Illustration of the scheme proposed by Mao et al. [30]

3.3.2 AoA based technique

Mao et al. have used AoA based technique for localization. A simple example for explanation of their concept is shown in Fig. 6.

Here M is the target node that needs to be located using a set of locator nodes. The locator nodes are installed with the necessary hardware to calculate AoA. As shown in Fig. 6, the locator nodes L_1 , L_2 and L_3 estimate the AoA as β_1 , β_2 and β_3 respectively at which the target node M is possibly located. The intersection of the lines of AoA of various locator nodes is the most likely location of the target node M . Under ideal circumstances the intersection should be a single point, however due to movement of the target node and the possible error in estimation of the AoA, a triangular region is formed. Typically, AoA estimation is done using an antenna array, wherein the phase difference

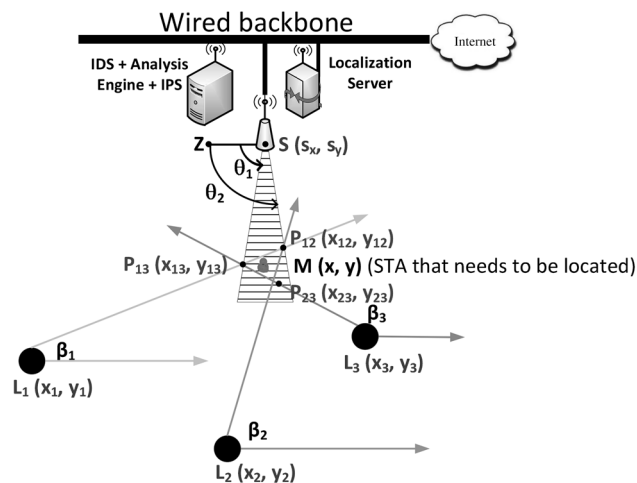


Fig. 7 Setup of the proposed localization system

between the received signals at each antenna array element is mapped to the incident direction of the signal. This method has two advantages compared to RSSI. First, since the phase of the received signal is usually more stable than the RSSI, AoA estimation can achieve higher accuracy than RSSI-based localization approaches. Second, given an effective AoA estimation scheme, the presence of two antenna arrays are sufficient to achieve accurate target localization, whereas range based approaches require three or more sensor elements [15]. Hence we use AoA based technique in our proposed IDS architecture in order to locate the attacker. To the best of our knowledge this technique has never been used in an IDS architecture.

- *Localization module for the IDS* The setup for the proposed AoA based Localization Module for the IDS is shown in Fig. 7. **M** is the location of the target STA that needs to be located, **S** is location of AP and **Z** is a reference point. The primary components that help in localization are the locator nodes and the Localization Server. The locator nodes communicate the location information of only authentication and association frames as they are responsible for flooding DoS attacks. The location of the locator nodes is known by the Localization Server before hand. The communication between the locator nodes and the Localization Server happens over Wi-Fi. The Localization Server and the AP communicate via dedicated wired backbone. The task of the locator node is to determine β_1 , β_2 and β_3 for the received authentication and association messages. The Localization Server uses β_1 , β_2 and β_3 in order to determine the location of the attacker (in a triangular region). Following that the Localization Server calculates angles θ_1 and θ_2 with respect to AP. The traffic from region $\theta_2 - \theta_1$ (shown in shaded lines

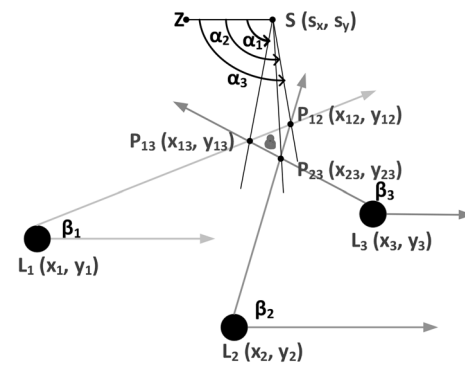


Fig. 8 Calculation of θ_1 and θ_2 from β_1 , β_2 and β_3 (in Fig. 7)

in Figure) is blocked since the target STA (**M**) that is flooding the network is located here. Now we explain in brief the procedure to compute θ_1 and θ_2 from β_1 , β_2 and β_3 using Fig. 8. Let the co-ordinates of the triangular region formed using the three locator nodes be denoted by $\{P_{13}, P_{12}, P_{23}\}$. Let $P_{13} = (x_{13}, y_{13})$ denote the intersection point of the AoA for the locator nodes L_1 and L_3 . Point P_{13} can be obtained as:

$$x_{13} = \frac{(y_3 - y_1) - (x_3 \cdot \tan \beta_3) + (x_1 \cdot \tan \beta_1)}{\tan \beta_1 - \tan \beta_3}$$

$$y_{13} = (x_{13} \cdot \tan \beta_1) + (y_1) - (x_1 \cdot \tan \beta_1)$$

On similar lines points P_{12} and P_{23} can be determined. Since co-ordinates of points **Z** (that is reference point) and **S** (that is AP) are known, the angle formed using the points **Z**, **S**, P_{13} can be determined. Let α_1 denote the angle **Z**, **S**, P_{13} , which can be calculated as:

$$\alpha_1 = \tan^{-1} \left(\frac{s_y - y_{13}}{s_x - x_{13}} \right)$$

On similar lines angles α_2 and α_3 can be determined. θ_1 and θ_2 can be derived as follows.

$$\theta_1 = \min(\alpha_1, \alpha_2, \alpha_3)$$

$$\theta_2 = \max(\alpha_1, \alpha_2, \alpha_3)$$

The derivation shown above is based on principles demonstrated in [22].

- *Blocking of traffic from the attacker region* After the AP receives the angles θ_1 and θ_2 and the co-ordinates of the STA from the Localization Server it checks if the MAC address in the received information corresponds to an attacker STA. If the MAC address is that of an attacker the AP blocks the traffic coming from the region corresponding to the angle $\theta_2 - \theta_1$ (shown in shaded region in Fig. 7) till the effect of flooding DoS attack subsides. The Localization Module induces additional frames in the network which increases the

load in the network. The consequence of this is discussed in Section.

3.4 Intrusion prevention system (IPS)

The *intrusion prevention system* module takes corrective actions once the flooding DoS attack is observed. The corrective action could include dis-connecting the clients, suspension of services offered by the AP etc. In the proposed approach the IDS first detects the occurrence of flooding DoS attack after which the localization module locates the attacker region and then the AP blocks the traffic coming from the attacker region only.

4 The design of intrusion detection system (IDS)

The dataset for flooding DoS attack was generated at Computer Science and Engineering Department of IIT Guwahati due to non-availability of any public dataset for flooding attacks. A set of 40 machines having wireless connectivity in them are connected to the departmental Wi-Fi network. Each machine was connected to one of the 5 APs available. We designate seven machines as attacker machines from which we launch the flooding DoS attacks on the Wi-Fi network. In order to generate an un-biased set of traces the traffic is captured on the normal institute network. This ensured that the traffic generated has a mixture of normal as well as attack activity. We captured the network traffic during different times of the day. Each trace consisted of a capture of 4 h of network traffic belonging to a particular time slot. Three slots are chosen during the day, first slot is the morning slot from 09:00–13:00 hours, the second slot is the afternoon slot from 14:00–18:00 hours, and the last is the evening slot from 20:00–00:00 hours. The traces are collected for a period of 1 week. The network traffic is usually heavy during the morning and the afternoon slots whereas it is relatively sparse during the evening slot. Also during the weekends it is observed that the morning and the evening slots have relatively less traffic as compared to the traffic these slots have during the weekdays. The flooding DoS attacks are launched multiple times during each slot using the aircrack-ng suite [2]. We also noted the timestamps during which we launched the flooding DoS attacks. This helped us in labeling the traffic correctly. We have used around 66 % of the dataset generated for training purposes and the remaining 34 % is used for testing purposes.

4.1 Feature selection for ML based IDS

For feature selection we have analyzed the communication between STA-AP pair under normal and attack scenarios

Table 1 Ranking of features

Weightage %	Feature
78.38	Transmission control protocol (TCP)
64.74	Inter frame distance (IFD)
60.46	Domain name server (DNS)
55.97	Address resolution protocol (ARP)
55.30	Association request
50.35	User datagram protocol (UDP)
48.63	Probe response
46.76	Authentication
39.63	Association response
36.87	Internet control message protocol (ICMP)
36.87	Null data/power save poll (PS-Poll)
26.49	De-authentication frames
21.59	NetBIOS datagram service (NBDS)
20.10	NetBIOS over TCP/IP (NBNS)
11.90	Dynamic host configuration protocol (DHCP)
7.88	Disassociation
7.88	Logical link control (LLC)
5.86	Probe request

from the dataset generated. For obtaining the STA-AP pair communication the entire data-set is split into individual AP data-set, which is further broken down into individual STA-AP pair communication. The behavior of the STA-AP pair is analyzed to determine whether the STA is in normal condition or anomalous. The number (#²) and the types of the frames that a STA exchanges with the AP is crucial in determining whether an attack has occurred or not. Using this information we have listed down 18 features in decreasing order of their significance as shown in Table 1. The significance is determined by Chi square test. We have specifically chosen only these 18 features since they play a vital role in the detection of flooding DoS attack. The lowest weighted feature as seen in Table 1 is Probe Request frame having a weight of only 5.86. All other features which are not used have weights zero, and consequently have no impact on the detection of flooding DoS attack.

The list of features along with their motivation behind selection for training the system for flooding DoS attack detection is given below. All the features except inter frame distance (IFD) are standard network frames.

1. # *Transmission control protocol (TCP) frames* TCP frames are present in good number in a normal network interaction of a legitimate user. However a malicious user launching flooding DoS attack creates

² # is used as a shorthand for *number of*. #TCP frames implies *number of* TCP frames.

large number of fictitious connections that hardly involve any TCP exchange. Less amount of the TCP frame exchange indicates towards flooding DoS attack.

2. *Inter frame distance (IFD)* The inter frame distance is the difference between the frame numbers of two identical frame types, as seen by the sniffer. For e.g., if the first and the second frame number of the authentication frame seen by the IDS are 626 and 1,098 respectively, then the IFD between the two authentication frames is $(1,098 - 626) = 472$. A series of such successful authentication with meager or no frame exchange is a pointer towards a suspected flooding attack impending on the network. Since many clients appear to join the targeted AP in a very short span of time under a flooding DoS attack, the IFD rapidly falls during that period. These activities provide vital evidences for the ML based IDS to detect the occurrence of the flooding attack. As against this, under normal circumstances, the client(s) do join and leave occasionally resulting in higher IFD.
3. *# Domain name server (DNS) frames* If no DNS frames are exchanged by STA with AP, it points towards fictitious connection with the AP. A high number of fictitious connections provides strong evidence of attack.
4. *# Address resolution protocol (ARP) frames* ARP frames are frequently exchanged among STAs to obtain MAC address of the destination. Lack of ARP frames provides a useful indicator for attack.
5. *# Association request frames* In normal circumstances only one association request frame is enough for associating a STA unless the frame is lost, in which case it is re-transmitted. Large count of association request frames is of paramount importance in indicating attack.
6. *# User datagram protocol (UDP) frames* The reason for inclusion of UDP is similar to those of TCP.
7. *# Authentication frames* Similar to Association Request frames.
8. *# Association response frames* In association flooding DoS attack AP often receives a large number of the spoofed association request frames. Absence of association response frame to a valid association request frame proves to be a critical factor in determining the presence of attack.
9. *# De-authentication frames* Large number of de-authentication frames are being sent by the AP to authentication requests, is an indicator towards attack activity.
10. *# Probe response frames* Similar to Association Response frames.
11. *# Internet control message protocol (ICMP) frames* Increase in number of ICMP error messages indicate towards flooding DoS attack.
12. *# Null data/power save poll frames* Legitimate STA generally frequently enters sleep state to conserve battery. Absence of null data or power save poll frames aids in detection of flooding DoS attack.
13. *# NetBIOS datagram service frames (NBNS)* During flooding DoS attack a large number of registrations are done in a small time windows.
14. *# NetBIOS over TCP/IP (NBNS)* Similar to DNS.
15. *# Dynamic host configuration protocol (DHCP) frames* A large number of DHCP requests are seen in a short span of time during attack.
16. *# Disassociate frames* In flooding DoS attack the AP sends dis-association frames to new association requests that arrive. Sending of large number of dis-association frames thus provides a useful discriminatory feature for attack.
17. *# Logical link control frames* The errors related to frame synchronization, flow control and error checking increase during attack.
18. *# Probe request frames* In flooding DoS attack since AP becomes un-responsive, probe requests are not honored. Hence the inclusion.

Table 2 depicts a snapshot of the statistics and the feature set table maintained by the Flood Analysis Module. Due to space constraints Table 2 is truncated and only some features are shown as a part of the training data. Table 2 is populated as follows: We take example of STA 4 shown in Table 2. At the start STA 4 has all counters 0. After sending an authentication request to the AP its Authentication count is incremented to 1. If the authentication is successful the AP responds with an authentication response with increments the Authentication count to 2. In this manner the counts for other features are updated. Based on the count of the various features the ML based IDS determines whether flooding DoS attack has occurred in the network. As an example, STA 10 exchanged 19 Authentication frames and 4 Association frames. This clearly indicates that STA 10 is flooding the AP with authentication and association requests. STA 10 has no TCP/UDP frame exchange and the IFD for STA 10 is 32 which is pretty low, which further proves that STA 10 is an attacker STA. STA 10 does no useful communication with the AP. It only injects a large number of authentication and association frames in the network. Hence this activity is rightly classified as an attack (A). On similar lines, if we see the behavior of STA 11 it exchanges only two authentication and two association frames (one each for request and response). Also the number of TCP and UDP frame exchanges are 136 and 52 respectively, which shows

Table 2 Features for classification of training data

STA	#TCP	IFD	#DNS	#ARP	#Association request	#UDP	#Authentication	#Association response	...	Result
STA 1	0	9	0	0	1	0	37	1	...	A
STA 2	0	16	0	0	0	0	14	1	...	A
STA 3	0	6	0	0	2	0	12	1	...	A
STA 4	35	495	10	10	1	20	2	1	...	N
STA 5	12	754	4	2	1	21	2	1	...	N
STA 6	0	94	0	0	0	0	12	1	...	A
STA 7	0	12	2	0	18	0	20	1	...	A
STA 8	110	451	7	1	1	10	3	1	...	N
STA 9	15	216	9	1	1	0	2	1	...	N
STA 10	0	32	0	0	4	0	19	1	...	A
STA 11	136	659	2	1	1	52	2	1	...	N
...	...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...
STA 12	70	1,506	4	5	1	46	2	1	...	N

that STA is actively communicating with the AP. The IFD of 659 is large enough. This corresponds to normal communication pattern of a STA and is classified as Normal(N).

4.2 Classifier design and selection

The success of ML based IDS depends heavily on the choice of a classifier. The job of a classifier is to accurately differentiate between attack and normal frames. In this section, we first describe few classification algorithms that are used in our proposed scheme. Each of the classification approach has its own pros and cons in terms of processing speed, accuracy and detection rate. From the perspective of an IDS accuracy and detection rate should be as high as possible. An administrator can choose amongst various techniques discussed below based on his network characteristics and requirements.

Classification algorithms are widely used in various applications. Data classification is a two step process. The first step is the training phase where the classification algorithm builds a classifier with the training data. The second step is classification wherein the classification model obtained in first step is used for classification and its performance is analyzed using test data. We describe a few classification techniques that we have used for our work.

4.2.1 Bayesian networks

Bayesian networks (BNs) or Bayes Nets belong to the family of probabilistic *graphical models* [17]. A BN is an annotated directed acyclic graph in which each node represents a random variable, while the edges between nodes represent the probabilistic dependencies among the

corresponding random variables. These conditional dependencies are estimated using known statistical and computational methods. The links between the nodes in BNs can be explained as correlation or association between random variables.

4.2.2 AdaBoosting.M1

AdaBoost short for *Adaptive Boosting* is a meta-heuristic [21]. In *AdaBoost* instead of attempting to generate a single complex prediction rule, training data is used to generate a large collection of very simple crude rules of thumb. A weight for each rule of thumb is then computed. A prediction about new input is made by combining the rules of thumb, taking into account each simple rule's weight and arriving at a consensus outcome. *AdaBoost* is adaptive in the sense that subsequent classifiers built are tweaked in favor of those instances misclassified by previous classifiers. *AdaBoost* is used in conjunction with many other learning algorithms to improve their performance. However, *AdaBoost* is sensitive to noisy data and outliers.

4.2.3 Alternating Decision Tree

The Alternating Decision Tree (ADTree) is a generalization of decision trees, voted decision trees and voted decision stumps [20]. The learning algorithm here is based on boosting. ADTree provides an extra layer of structure to the set of weak learners derived in the boosting algorithm. An ADTree consists of decision nodes and prediction nodes. *Decision nodes* specify a predicate condition. *Prediction nodes* contain a single number. An instance is classified by following all the paths for which all *decision nodes* are true and summing any *prediction nodes* that are traversed. The

generated classification rules are often smaller and can be easily interpreted.

4.2.4 SVM

Support Vector Machines or SVMs belongs to class of classifiers that simultaneously minimize the empirical classification error and maximize the geometric margin. The process involves creating a hyperplane in N-dimensional space that would separate two data sets with highest margin [18]. SVMs classify data by determining a set of support vectors which are members of the set of training inputs that outline a hyper plane in the feature space. SVMs through the use of a kernel function provides a mechanism to fit the surface of the hyperplane to the data. The user may provide various functions (example, linear, polynomial, or sigmoid) to the SVMs during the training process, which selects support vectors along the surface of this function.

4.2.5 RIDOR

Ripple-Down Rule Learner or RIDOR like all rule learners induces a set of rules from the data. It generates the default rule first and then the exceptions for the default rule with the least (weighted) error rate. The process is repeated until the final leaf is reached which has only one default class and no exceptions.

5 Experimental setup and results

The test-bed setup consists of a NETGEAR AP with SSID “WiFi-Free” along with an IDS infrastructure placed as shown in Fig. 9. The IDS is placed just beside the AP to ensure that it captures maximum frames coming towards or leaving the AP since in flooding DoS attacks APs are the primary target. Attacker STA is loaded with BackTrack 5R3 [3]. BackTrack operating system consists of the *air-crack-ng* suite which is used to launch Authentication and Association flooding DoS attacks. The attacker’s prime target is to overwhelm the victim AP with large number of authentication and association frames so that the victim AP is unable to provide services to legitimate clients.

It is assumed that the attacker will try to evade detection. An attacker may slow down the frame injection rate in order to escape detection. However lowering the frame injection rate diminishes the effect of the attack and the network behavior is similar to normal network conditions. Thus, lowering the frame injection rate may help the attacker evade detection but the legitimate user’s service is not degraded, defeating the purpose of flooding DoS

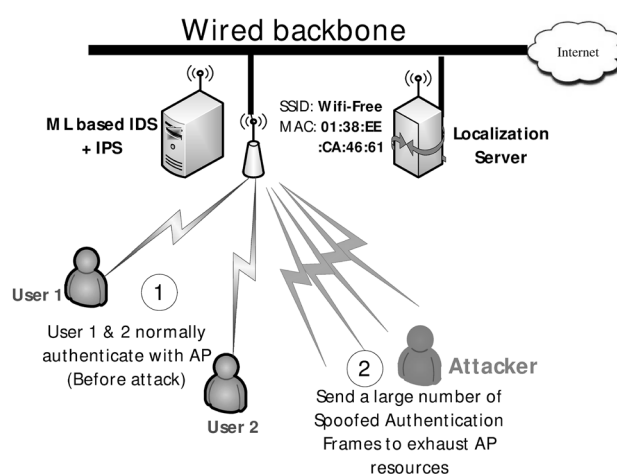


Fig. 9 Experimental setup

attacks. Hence the attacker will always aim to inject frames at a higher rate in-order to launch flooding DoS attacks.

In the following sub-sections we describe the generation of training and testing data along with feature selection for the attacks under consideration.

5.1 Accuracy and detection rate of proposed IDS

The metrics used for measuring the performance of IDS are accuracy (i.e., Precision) and detection rate (i.e., Recall).

Accuracy is the proportion of the total number of attacks that are correctly detected. It is determined using the equation:

$$Accuracy = Precision = \frac{TP}{TP + FP}$$

Detection Rate is defined as the number of attacks detected by the IDS to the total number of attacks actually present.

$$DetectionRate = Recall = \frac{TP}{TP + FN}$$

Here, TP is True Positive, FP is False Positive, FN is False Negative. A TP is a case, which is actually an attack and is declared as attack by the IDS. A FP arises when IDS treats a normal activity as attack activity. A FN occurs when the IDS treats an attack activity as normal.

The CPU and memory consumption are vital parameters when considering the scalability of IDS. However these parameters are secondary parameters and have no effect on the detection rate and accuracy. The CPU and memory consumption of an IDS are issues related to implementation. Hence in this work we only concentrate on accuracy and detection rate of the ML based IDS [35].

We have compared the performance in terms of accuracy and detection rate for the various classification techniques built in the WEKA [24] tool for the data-set generated.

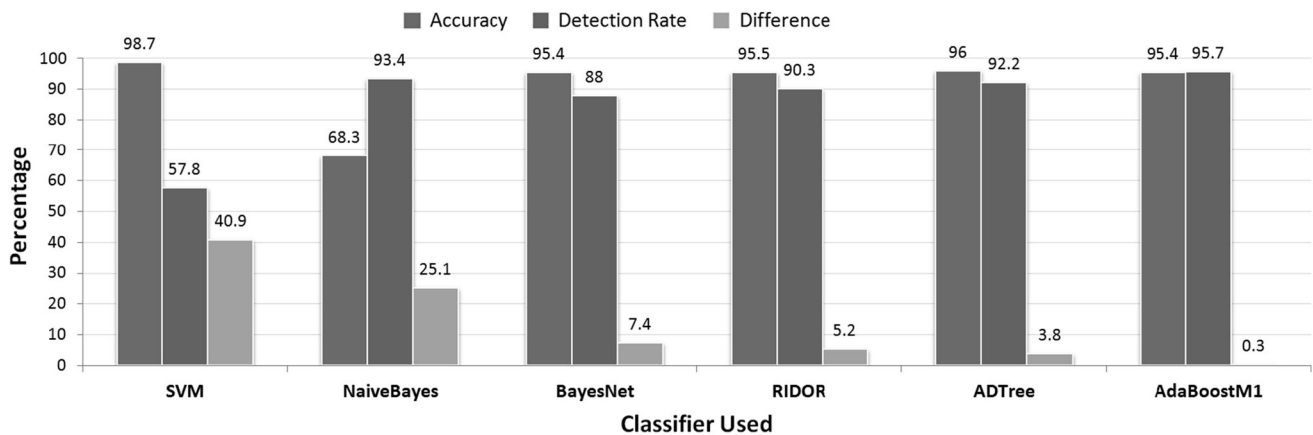


Fig. 10 Attack detection rate, accuracy and the absolute difference between them of the classifiers used

From the graph shown in Fig. 10 we see that various classes of classifiers yield promising results in terms of detection rate and accuracy of flooding DoS attacks. The purpose of choosing the various classifiers amongst different classes is to enable the administrator for selection of a suitable classification algorithm, based on the network characteristics. Naive Bayes has the weakest performance overall in terms of accuracy (only 68 %) though the detection rate is quite good. Naive Bayes requires the least time to build the model among all the classifiers used. It required only 2.02 s to build the model. Bayes Net which is another probabilistic classifier but has significant improvement over Naive Bayes. The accuracy and detection rate for BayesNet is 95 and 88 %, respectively. However the time taken to build the model in Bayes Net is higher than Naive Bayes, but it still second best when comparing the build times of the classifiers used. Support Vector Machines (SVM) has the best accuracy rate of 98.7 % but has the lowest detection rate of 57.8 % which is abysmally low from an IDS perspective. Its has the worst model building time of 208.25 s among the classifiers used. RIDOR is a rule based classifier that has both accuracy and detection rate above 90 %. Its accuracy is 95.5 % while the detection rate stands at 90 %. Still the difference of 5 % between the accuracy and detection rate is quite high. ADTree is a tree based algorithm possess an accuracy of 96 % and has a detection rate of 92.2 %. ADTree is one of the best performing algorithm for the detection of flooding DoS attacks. The model build time of RIDOR and ADTree are comparable as both took close to 75 s to build the model. AdaBoost is a boosting technique that has the best accuracy and detection rate for the flooding DoS attacks. Though its accuracy is slightly less than ADTree (less by a margin of 0.6 %), the difference between the detection rate and accuracy is a mere 0.3 %. The model building time is 26.41 s which is more than Naive Bayes and BayesNet, but significantly lesser than SVM, RIDOR and AdaBoost. With

Table 3 Comparison of various classification techniques used for detection of flooding DoS attacks

Classifier	Accuracy (precision)	Detection rate (recall)	F-measure	ROC area	Time taken to build model (s)
Naive Bayes	0.683	0.934	0.789	0.85	2.02
BayesNet	0.954	0.88	0.915	0.97	8.56
SVM	0.987	0.578	0.729	0.785	208.25
Ridor	0.955	0.903	0.928	0.93	73.98
ADTree	0.96	0.922	0.941	0.988	74.55
AdaBoostM1	0.954	0.957	0.956	0.98	26.41

both detection rate and accuracy and detection rate more than 95 %, AdaBoost certainly is the best choice among various classifications algorithms we have used.

We first compare the two Bayes Classifier algorithms. The first one is the Naive Bayes and the other is BayesNet. As we see in Table 3 Naive Bayes's accuracy is mere 68.3% while its detection rate is 93.4 %. With an accuracy of 95.4 % BayesNet classifier has a significant improvements over Naive Bayes in terms accuracy, however its detection rate of 88 % is slightly below Naive Bayes detection rate. The primary reason behind the improvement of BayesNet classifier technique is due to the fact that Naive Bayes assumes all the features to be *independent* of one another. BayesNet on the other hand does consider joint probabilities among the features. The network traffic is seldom independent. For example, when a genuine STA successfully associates with an AP it implies that the STA must have successfully authenticated. It is not possible for a STA to associate successfully after un-successful authentication. Thus, a successful association does *depend* on successful authentication. Similarly, exchange of TCP and UDP frames are inter-related. BayesNet considers such

joint probabilities and does not assume the features to be independent like Naive Bayes. Hence results obtained using BayesNet are better than those obtained for Naive Bayes.

Though RIDOR (Ripple-Down Rule learner) has an accuracy of 95.5% and its detection rate is 90 %. Rule based machine learning algorithms generally require verification by security expert for the rules generated. Human intervention is susceptible to error since different experts may have different view points in labeling a particular rule as attack or normal. If the rules are chosen improperly, it would have an adverse effect the outcome of the results.

AdaBoost (with Decision stumps) has an accuracy of 95.7 % while the detection rate being 95.4 %. For the problems which tend to have varying degree of hardness boosting provides very good results. Boosting algorithm tends to generate distributions that concentrate on the harder examples which challenge the weak learning algorithm to perform well on these harder parts of the sample space. Also boosting tries actively to force the weak learning algorithm to change its hypotheses by constructing a “hard” distribution over the examples based on the performance of previously generated hypotheses (variance reduction). AdaBoost significantly reduces the error of the learning algorithm, yielding better results. AdaBoost model build time though more than Naive Bayes and BayesNet but is significantly less when compared with likes of SVM and ADTree. The high detection rate and accuracy of AdaBoost classifier as compared to Naive Bayes and Bayes Net compensates for the higher build time.

Support Vector Machines (SVMs) with an accuracy of 98.7 % stands out amongst all the classification algorithms discussed in this paper. However, the detection rate of SVM is the lowest among the classifiers used. With an attack detection rate of 57.8 % it implies that for every 100 attacks, 42 attacks are not reported. The primary reason for this is the imbalanced nature of the dataset. SVMs generally do not perform well on highly skewed or imbalanced data sets. In our scenarios even though the attacks are launched frequently, the traces obtained for normal traffic pattern are comparatively more than the traces obtained for the attack scenario. Again, the accuracy and detection rate in SVM are highly influenced by the choice of kernel. The network data being heavily heterogeneous selection of an appropriate kernel that performs well under all network conditions is another issue. From a network administrator point of view the complex data transformations and the resulting boundary plane obtained in SVM classification technique are very difficult to interpret. In comparison, it is very easy to interpret the results obtained using decision trees algorithms. ADTree is a combination of decision trees and ADABOOST classification algorithms. ADTree produces

an accuracy of 96 % while the detection rate stands at 92.2 %. In comparison to SVMs, the ADTree significantly improves in terms of detection rate at the expense of slight fall in the accuracy. SVMs also suffer from high algorithmic complexity and extensive memory requirements. In the experiments conducted ADTree’s model build time (74.55) is almost three times faster than SVMs (208.25). In addition to classification ADTree also gives measure of confidence in the prediction. These reasons add to the performance of ADTrees.

The proposed ML based IDS performs really well for a host of classification algorithms. In summary, AdaBoost technique performs the best in terms of detection rate and accuracy (i.e., both exceed 95 % mark) and also has low model build time.

5.2 Localization

The testbed shown in Fig. 9 is small for evaluating the results for localization algorithms. So we have simulated the proposed localization techniques. The scenario for evaluating the localization component is shown in Fig. 11. L1, L2, ..., L9 are the locator nodes. The locator nodes are spread within an area 8×8 m area and the attacker is placed at various locations. All the units for measurement are in meters. During the simulation attacker is placed at various locations ranging from A, B, ..., J. The experimental results for various localization algorithms are demonstrated in Table along with their mean distance error. The Table 4 clearly indicates that the AoA based approach performs significantly better than RSSI based approach.

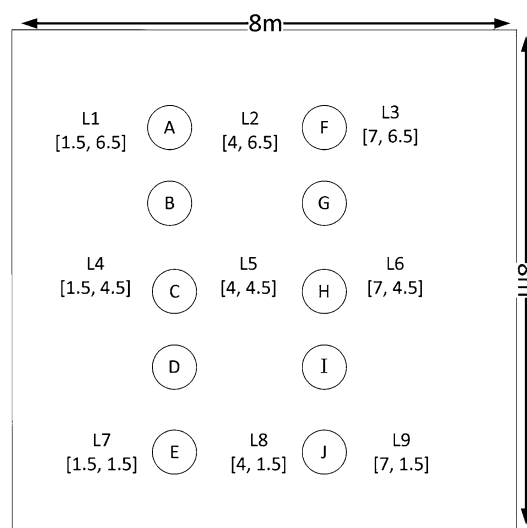


Fig. 11 Simulation setup for localization

5.2.1 Network load due to localization

Considering only authentication and association frames are used for the flooding DoS attacks, AoA for the frames other than authentication and association are not computed by the locator nodes. The localization technique requires that, for every authentication/association frame received by the locator nodes, at-least three locator nodes forward the location messages to the Localization Server. In addition to this the Localization Server computes θ_1 and θ_2 with respect to AP and the co-ordinates of the attacker, and forwards it to AP. So, for every authentication/association frame transmitted in the network three additional frames are required in order to locate the attacker.

Table 4 Experimental results for localization using AoA, RSSI and the hybrid methodology

Attacker node ID	Actual co-ordinates of attacker (x, y)		Calculated co-ordinates of attacker (x, y)		Error	Calculated co-ordinates of attacker (x, y)		Error
			AoA based			RSSI based		
A	2.75	6.5	3.12	6.74	0.441	3.21	6.84	0.572
B	2.75	5.5	3.08	5.64	0.358	2.14	5.73	0.652
C	2.75	4.5	3.16	4.71	0.461	2.27	4.82	0.577
D	2.75	3	3.22	3.32	0.569	3.24	3.36	0.608
E	2.75	1.5	3.14	1.67	0.425	3.26	1.74	0.564
F	5.5	6.5	5.82	6.69	0.372	5.91	6.79	0.502
G	5.5	5.5	5.91	5.82	0.520	6.02	5.99	0.714
H	5.5	4.5	5.63	4.74	0.273	5.6	4.82	0.335
I	5.5	3	5.74	3.29	0.376	5.72	3.45	0.501
J	5.5	1.5	5.81	1.74	0.392	5.84	1.78	0.440
Average distance error					0.419	0.547		

Under normal network circumstances the load on the network is relatively low. As a result of this, the additional three frames for every authentication/association frame does not induce any significant increase in the number of frames transmitted. When flooding DoS attacks are launched the network is already overloaded and an additional three frames per authentication/association frame further increases the number of frames in the network. This is evidently visible in the graph for the network traffic shown in Fig. 12. The peaks in Fig. 12 denote the region during which the flooding DoS attacks are observed in the network. As we see in Fig. 12 during the flooding DoS attacks, an appreciable difference for the number of frames between normal and attack scenario is observed. However, if the attacker injects a large number of authentication and associations frames targeting a particular STA, then the locator nodes send the localization information only once. This helps in conserving the network bandwidth. Figure 12 shows that most often the network traffic with and without localization module are almost the same.

In the proposed ML based IDS, the IDS blocks the traffic coming from the attacker region once the attacker is located. As a result the network recovers quickly from flooding DoS attack and the services of the non-victim clients are not affected.

6 Conclusion

In this paper, we have proposed a novel Machine Learning based Intrusion Detection System for Flooding DoS attacks in 802.11 networks. The proposed ML based IDS detects the flooding DoS attacks with high detection rate and low false positive rate for AdaBoost, BayesNet, RIDOR and ADTree classifiers. The AdaBoost algorithm has both the accuracy and detection rate above 95%. Another major advantage of the ML based IDS is that it does not require

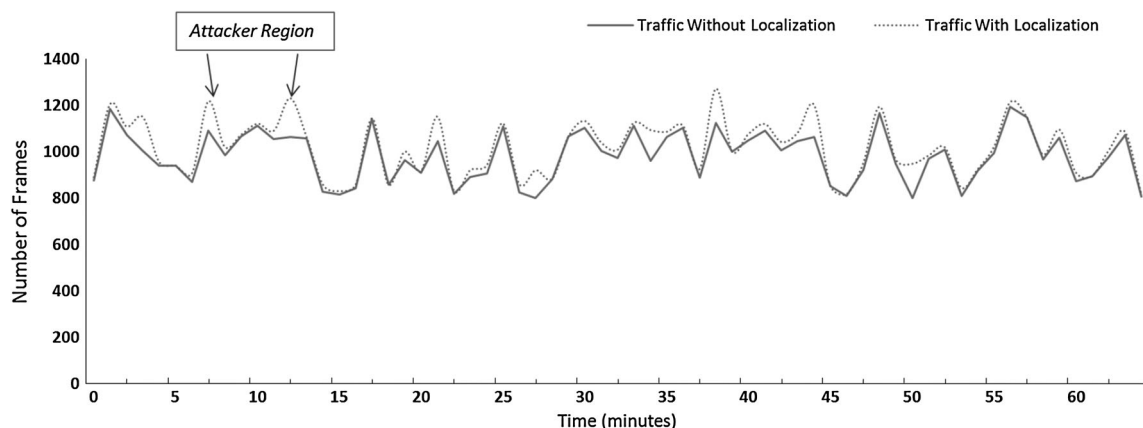


Fig. 12 Comparison of network traffic without and with the localization module

protocol modifications, use of any encryption algorithms or firmware upgrades either on the AP or the end users. Besides this, the proposed work can be applied on legacy as well as present-day systems. As a part of future work, we would like to test the efficacy of ML based techniques on other MAC layer attacks occurring in 802.11 Wi-Fi networks. Currently we have used 18 features for identifying the occurrence of flooding attacks. We would like to reduce the number of features used for identifying the flooding attacks. In this respect we would like to explore more recent machine learning classifiers and feature selection algorithms. The reduced number of features will also curtail the learning time required by the classifiers. We have used the default values of the parameters of various classifiers used in Weka. We would also like to tweak these default values and analyze its effect on detection rate and accuracy.

After detection of an attack the proposed scheme also locates the attacker using an AoA based technique. All traffic originating from the region of the attacker is blocked so that effect of attack is subsided and the legitimate users continue to use the services offered the AP in an uninterrupted manner. However, the AoA parameter is heavily influenced by the surroundings and the strength gets effected in presence of obstacles like walls, buildings etc. thereby introducing errors while estimating location. In order to reduce this error, locator nodes can be trained to estimate the distance of STA more accurately in the given surroundings. Also, the localization module sends at-least three frames for every authentication and association frame which increases network activity. We would like to test the efficacy of distributed algorithms for the localization module in-order to reduce the number of frames transmitted by the localization module. The use of distributed algorithms can help to reduce this number which would bring down the network load.

Acknowledgments The first author of this paper is supported by TATA Consultancy Services (TCS), India, through TCS Research Fellowship Program. We also acknowledge Dr. Vijaya Saradhi and Dr. Sanasam Ranbir Singh, Assistant Professor, Department of Computer Science and Engineering, IIT Guwahati, for their constructive suggestions and helpful insights in dealing with problems relating to Machine Learning algorithms. We would like to also acknowledge Sandip Chakraborty, Research Scholar, IIT Guwahati, for helping us with the queries related to localization module.

References

1. Grimwepa—WEP and WPA Password Cracker. <http://code.google.com/p/grimwepa/>
2. Aircrack-ng Suite. <http://www.aircrack-ng.org/>
3. BackTrack. <http://www.backtrack-linux.org/>
4. Detecting Wireless LAN MAC Address Spoofing Joshua Wright. <http://www.willhackforsushi.com/papers/wlan-mac-spoof.pdf>
5. File2air. <http://www.willhackforsushi.com/File2air.html>
6. Scapy. <http://www.secdev.org/projects/scapy/>
7. WEPCrack—An 802.11 key breaker. <http://wepcrack.sourceforge.net/>
8. Wireshark. <http://www.wireshark.org>
9. IEEE Standard for information technology-Telecommunications and information exchange between systems-Local and metropolitan area networks-Specific requirements—Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications. IEEE Std 802.11-2007 (Revision of IEEE Std 802.11-1999) pp C1–1184 (2007)
10. IEEE Standard for information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements. Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications. Amendment 4: Protected Management Frames. IEEE Std. 802.11w-2009, (Amendment 4: Protected Management Frames) pp 1–111 (2009)
11. Anjum F, Das S, Gopalakrishnan P, Kant L, Kim B (2005) Security in an insecure WLAN network. In: 2005 International Conference on Wireless Networks, Communications and Mobile Computing, pp 292–297
12. Bellardo J, Savage S (2003) Denial-of-Service Attacks: Real Vulnerabilities and Practical Solutions. In: Proceedings of the 12th Conference on USENIX Security Symposium—vol 12, SSYM'03, 802.11, pp 2–2
13. Bernaschi M, Ferreri F, Valcamonici L (2008) Access points vulnerabilities to DoS attacks in 802.11 networks. *Wirel Netw* 14(2):159–169
14. Bittau A, Handley M, Lackey J (2006) The Final Nail in WEP's Coffin. In: Proceedings of the 2006 IEEE Symposium on Security and Privacy., SP '06IEEE Computer Society, Washington, DC, USA, pp 386–400
15. Chen HC, Lin TH, Kung H, Lin CK, Gwon Y (2012) Determining RF Angle of Arrival using COTS antenna arrays: a field evaluation. In: Military Communications Conference, 2012—MILCOM 2012, pp 1–6
16. Cheng CT, Lin JY, Sun YG, Chau K (2005) Long-term prediction of discharges in manwan hydropower using adaptive-network-based fuzzy inference systems models. *Adv Nat Comput Lect Notes Comput Sci* 3612:1152–1161
17. Cleophas T, Zwinderman A (2013) Bayesian Networks. In: Machine Learning in Medicine. Springer, Netherlands, pp 163–170
18. Doukas C, Maglogiannis I, Tragas P, Liapis D, Yovanof G (2007) Patient Fall Detection using Support Vector Machines. In: Boukis C, Pnevmatikakis A, Polymenakos L (eds) Artificial Intelligence and Innovations 2007: from theory to applications, IFIP The International Federation for Information Processing, vol 247. Springer, US, pp 147–156
19. Faria DB, Cheriton DR (2006) Detection of identity-based attacks in wireless sensor networks using signalprints. In: Proceedings of the 5th ACM workshop on Wireless security, WiSe '06, pp 43–52
20. Freund Y, Mason L (1999) The alternating decision tree learning algorithm. In: Proceedings of the Sixteenth International Conference on Machine Learning, ICML '99, pp 124–133
21. Freund Y, Schapire RE (1996) Experiments with a new boosting algorithm. In: International Conference on Machine Learning, pp 148–156
22. Gavish M, Weiss A (1992) Performance analysis of bearing-only target location algorithms. *IEEE Trans Aerosp Electron Syst* 28(3):817–828
23. Guo F, Chiueh TC (2006) Sequence number-based MAC address spoof detection. In: Proceedings of the 8th International Conference on Recent Advances in Intrusion Detection, RAID'05, pp 309–329

24. Hall M, Frank E, Holmes G, Pfahringer B, Reutemann P, Witten IH (2009) The WEKA Data Mining Software: an update. *SIG-KDD Explor* 11(1):10–18
25. LaRoche P, Zincir-Heywood A (2006) De-authentication attack detection using genetic programming. In: *Genetic Programming*, vol 3905, 802.11, pp 1–12
26. LaRoche P, Zincir-Heywood AN (2006) Genetic programming based WiFi data link layer attack detection. In: *Proceedings of the 4th Annual Communication Networks and Services Research Conference, CNSR '06*, pp 285–292
27. Liu D, Ning P, Du W (2005) Attack-resistant location estimation in sensor networks. In: *Fourth International Symposium on Information Processing in Sensor Networks, 2005. IPSN 2005*, pp 99–106
28. Liu Y, Tian D, Li B (2006) A wireless intrusion detection method based on dynamic growing neural network. In: *Proceedings of the First International Multi-Symposiums on Computer and Computational Sciences—vol 2 (IMSCCS'06)*, IMSCCS '06, pp 611–615
29. Liu YH, Tian DX, Wei D (2006) A wireless intrusion detection method based on neural network. In: *Proceedings of the 2Nd IASTED International Conference on Advances in Computer Science and Technology, ACST'06*, pp 207–211
30. Mao G, Bariş F, Brian A (2007) Wireless sensor network localization techniques. *Comput Netw* 51(10):2529–2553
31. Mar J, Yeh YC, Hsiao IF (2010) An ANFIS-IDS against deauthentication DOS attacks for a WLAN. In: *International Symposium on Information Theory and its Applications (ISITA)*, pp 548–553
32. Martinovic I, Zdarsky FA, Schmitt JB (2007) Regional-based authentication against DoS attacks in Wireless networks. In: *Proceedings of the 3rd ACM workshop on QoS and security for Wireless and mobile networks, ACM*, pp 176–179
33. Ming Z, Wang H, Xu M, Pan D (2014) Efficient handover in railway networking via named data. *Int J Mach Learn Cybern* 5:1–7. doi:[10.1007/s13042-014-0282-9](https://doi.org/10.1007/s13042-014-0282-9)
34. Ming Z, Wang H, Xu M, Pan D (2014) Evaluation of path stretch in scalable routing system. *Int J Mach Learn Cybern* 5:1–7
35. Puketza NJ, Zhang K, Chung M, Mukherjee B, Olsson RA (1996) A methodology for testing intrusion detection systems. *IEEE Trans Softw Eng* 22(10):719–729
36. Stubblefield A, Ioannidis J, Rubin AD (2004) A key recovery attack on the 802.11b Wired Equivalent Privacy Protocol (WEP). *ACM Trans Inf Syst Secur* 7(2):319–332
37. Tews E, Beck M (2009) Practical attacks against WEP and WPA. In: *Proceedings of the Second ACM Conference on Wireless Network Security, WiSec '09*, pp 79–86
38. Tews E, Weinmann RP, Pyshkin A (2007) Breaking 104 Bit WEP in less than 60 seconds. *Inf Secur Appl Lect Notes Comput Sci* 4867:188–202
39. Wang WC, Cheng CT, Chau KW, Xu DM (2012) Calibration of Xinjiang model parameters using hybrid genetic algorithm based fuzzy optimal model. *J Hydroinform* 14:784–799
40. Wu CL, Chau KW, Li YS (2009) Predicting monthly streamflow using data-driven models coupled with data-preprocessing techniques. *Water Resour Res* 45(8):1–23
41. Xia H, Brustoloni J (2004) Detecting and Blocking Unauthorized Access in Wi-Fi Networks. In: *Networking Technologies, Services, and Protocols; Performance of Computer and Communication Networks; Mobile and Wireless Communications*, vol 3042, pp 795–806
42. Zhang J, Chau KW (2009) Multilayer ensemble pruning via novel multi-sub-swarm particle swarm optimization. *J Univers Comput Sci* 15(4):840–858